Experiment 2

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1. **AIM:** To implement Single Page Application (SPA) navigation in the EcoTrack application using React Router, secure application routes using context-based authentication, and extend nested dashboard routing through follow-up enhancements.
2. **Objective:**
   a. Configure client-side routing in a React application using React Router
   b. Implement SPA navigation without full page reloads
   c. Design and apply protected routes using route-guard patterns
   d. Manage shared authentication state using React Context API
   e. Implement nested routing to build dashboard-style layouts
   f. Extend existing nested routes by adding new dashboard sections
   g. Implement logout functionality by updating shared context state
   h. Analyze route access behavior and explain redirection logic.
   i. Understand the role of Context API in shared state management and its c omparison with Redux at an introductory level
3. **Implementation/Code:**
   a. Dashboard.jsx:

```jsx
import { Link, Outlet } from "react-router-dom";
const Dashboard = () => {
  return (
    <>
      <h2>EcoTrack Dashboard</h2>
      <nav>
        <Link to="profile">Profile</Link>{" | "}
        <Link to="reports">Reports</Link>
      </nav>
      <Outlet />
    </>
  );
};
export default Dashboard;
```

b. **Index.js:**

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(<App />);
```

c. AuthContext.js

```
import { createContext, useState } from "react";
export const AuthContext = createContext();
export const AuthProvider = ({ children }) => {
  const [isAuthenticated, setIsAuthenticated] = useState(false);

  const login = () => setIsAuthenticated(true);
  const logout = () => setIsAuthenticated(false);

  return (
    <AuthContext.Provider value={{ isAuthenticated, login, logout }}>
      {children}
    </AuthContext.Provider>
  );
};
```

4. **Output:**

Home | Dashboard | Login

Welcome to EcoTrack

Home | Dashboard | Login

Login

Login

Home | Dashboard | Logout

EcoTrack Dashboard

Profile | Reports

Home | Dashboard | Logout

EcoTrack Dashboard

Profile | Reports

Environmental Reports

5. **Learning Outcome:**
   a. I learnt how to make react routes.
   b. How the react routes actually works.
   c. I learnt how to use the outlet tags along with nested routes.